O-I-556C
February 5, 1974
SUPERSEDING
Fed. Spec. O-I-556b
September 24, 1963

FEDERAL SPECIFICATION

INSECTICIDE, METHYL BROMIDE

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers two types of methyl bromide insecticide.

1.2 Classification. Insecticide, methyl bromide shall be of the following types as specified (see 6.2):

Type I - Odorized.
Type II - Nonodorized.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

Federal Specifications:

RR-C-901 - Cylinders, Compressed Gas: With Valve or Plug and Cap; ICC 3AA.
PPP-B-585 - Boxes, Wood, Wirebound.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-C-96 - Cans, Metal, 28 Gage and Lighter.
PPP-F-320 - Fiberboard; Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes.

FSC 6840

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Federal Standard:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specifications:

MIL-C-10664 - Chloropicrin, Technical.
MIL-P-15011 - Pallet, Material Handling, Wood, Post Construction, 4-Way Entry.

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-147 - Palletized and Containerized Unit Loads, 40" x 48" Pallets, Skids, Runners, or Pallet-Type Base.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Laws and Regulations:

Federal Environmental Pesticide Control Act.

(The Federal Environmental Pesticide Control Act of 1972 is under the jurisdiction of the Office of Pesticides Programs, Environmental Protection Agency, Washington, DC 20250.)

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Code of Federal Regulations:

49 CFR 171-179 - Department of Transportation Rules and Regulations for the Transportation of Explosives and Other Dangerous Articles.

(The Department of Transportation regulations are a part of the Code of Federal Regulations available from the Superintendent of Documents, U. S. Government Printing Office, Washington, DC 20402. Orders for the above publication should cite "49 CFR 171-179.")

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

Uniform Classification Committee:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

National Classification Board:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., Tariff Order Section, 1616 P Street, N. W., Washington, DC 20036.)

American Society for Testing and Materials (ASTM) Standards:

D1193-70 - Reagent Water.
D1533-61 - Water in Insulating Liquids (Karl Fischer Method).
E1-71 - ASTM Thermometers.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

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3. REQUIREMENTS

3.1 Acidity (as hydrogen bromide). Insecticide, methyl bromide shall contain no more than 0.001 percent by weight acidity when tested as specified in 4.2.4.1.

3.2 Moisture. Insecticide, methyl bromide shall contain no more than 0.01 percent by weight moisture when tested as specified in 4.2.4.2.

3.3 Distillation characteristics. Insecticide, methyl bromide shall have a distillation range for the 1 to 95 percent fraction of no more than 0.6° C and an initial boiling point of no less than 3.2° C when tested as specified in 4.2.4.3.

3.4 Odorant. Type I insecticide, methyl bromide shall contain 2.0 ± 0.5 percent by weight chloropicrin conforming to MIL-C-10664. The chloropicrin shall be added after the insecticide, methyl bromide has passed all tests.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to specified requirements.

4.2 Quality conformance inspection.

4.2.1 Lotting. A lot shall consist of the insecticide, methyl bromide produced by one manufacturer, at one plant, from the same materials, and under essentially the same manufacturing conditions provided the operation is continuous. In the event the process is a batch operation, each batch shall constitute a lot (see 6.3).

4.2.2 Sampling.

4.2.2.1 For examination of preparation for delivery. Sampling for examination of preparation for delivery shall be conducted in accordance with MIL-STD-105.

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4.2.2.2 For test.

(a) Type I insecticide. Three representative specimens of approximately 250 milliliters (ml) each (volume when liquid) shall be taken from the lot of methyl bromide prior to the addition of the chloropicrin and placed in separate suitable clean, dry containers labeled to identify the lot from which they were taken.

(b) Type II insecticide. Sampling for test of type II insecticide shall be conducted in accordance with table I. A 250-ml specimen (volume when liquid) shall be removed from each sample container and placed in a suitable clean, dry container labeled to identify the lot and container from which it was taken.

Table I. Sampling for test of type II insecticide

Number of containers in batch or lot	Number of sample containers
2 to 25	2
26 to 150	3
151 to 1,200	5
1,201 to 7,000	8
Over 7,000	13

(c) Container leakage. Sampling for container leakage test shall be conducted in accordance with MIL-STD-105, inspection level III, using an AQL of 0.65 percent defective.

4.2.3 Inspection procedure.

4.2.3.1 For examination of preparation for delivery. The sample unit shall be one filled unit container or shipping container, as applicable. Sample unit and shipping containers and the preparation for delivery thereof shall be examined for the following defects using an AQL of 1.5 percent defective:

(a) Contents per container not as specified
(b) Container not as specified
(c) Container closure not as specified
(d) Container damaged
(e) Marking incorrect, missing, or illegible
(f) Palletization not as specified (when required)

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4.2.3.2 For container leakage test. The sample unit shall be one filled unit container. Sample unit containers taken as specified in 4.2.2.2(c) shall be tested for leakage as specified in 4.2.4.4 using an AQL of 0.65 percent defective.

4.2.3.3 For test. Each specimen taken in 4.2.2.2(a) or 4.2.2.2(b) shall be tested separately as specified in 4.2.4.1, 4.2.4.2, and 4.2.4.3. Failure of any test by any specimen shall be cause for rejection of the lot represented.

4.2.4 Tests. Water in accordance with ASTM D1193 and reagent grade chemicals shall be used throughout the tests. Where applicable, blank determinations shall be run and corrections applied where significant. Tests shall be conducted as follows:

4.2.4.1 Acidity. Place approximately 10 ml of a slurry of crushed ice and water into a 250-ml glass-stoppered Erlenmeyer flask. Add 50 ml of the specimen at 0° C to the slurry, stopper the flask loosely, and shake gently until the ice is completely melted. Remove the stopper and evaporate the specimen under a hood until practically all the methyl bromide is vaporized. Add 1 drop 0.10-percent methyl red indicator. If an orange or pink color develops, titrate with 0.01N sodium hydroxide solution to a clear yellow color. Calculate the percent by weight acidity as hydrogen bromide as follows:

$$\text{Percent acidity} = 0.0934AB$$

where: A = Milliliters of sodium hydroxide solution used in the titration and
B = Normality of the sodium hydroxide solution.

4.2.4.2 Moisture. Determine the percent by weight of moisture in the specimen in accordance with ASTM D1533.

4.2.4.3 Distillation characteristics.

(a) Apparatus. The apparatus as shown in figure 1 consists of a 125-ml Engler distilling flask to the bottom of which is attached a glass finger 1 centimeter (cm) in diameter and 3 cm in length. This is wound with 60 cm of Hoskins Chromel "A" BS gage No. 28 resistance wire and serves as the heating element. The distilling flask is almost completely surrounded by a dry ice atmosphere. The dry ice holder consists of a circular piece of 1/8-inch gasket material, 14 cm in diameter with a 2-cm hole in the center, around which is wrapped a wire mesh 7 cm high and 44 cm long. Inside and forming a concentric circle is another piece of wire mesh 6 cm by 26 cm which is also attached to the gasket material. These two circles of wire mesh are covered with aluminum foil and the space between them

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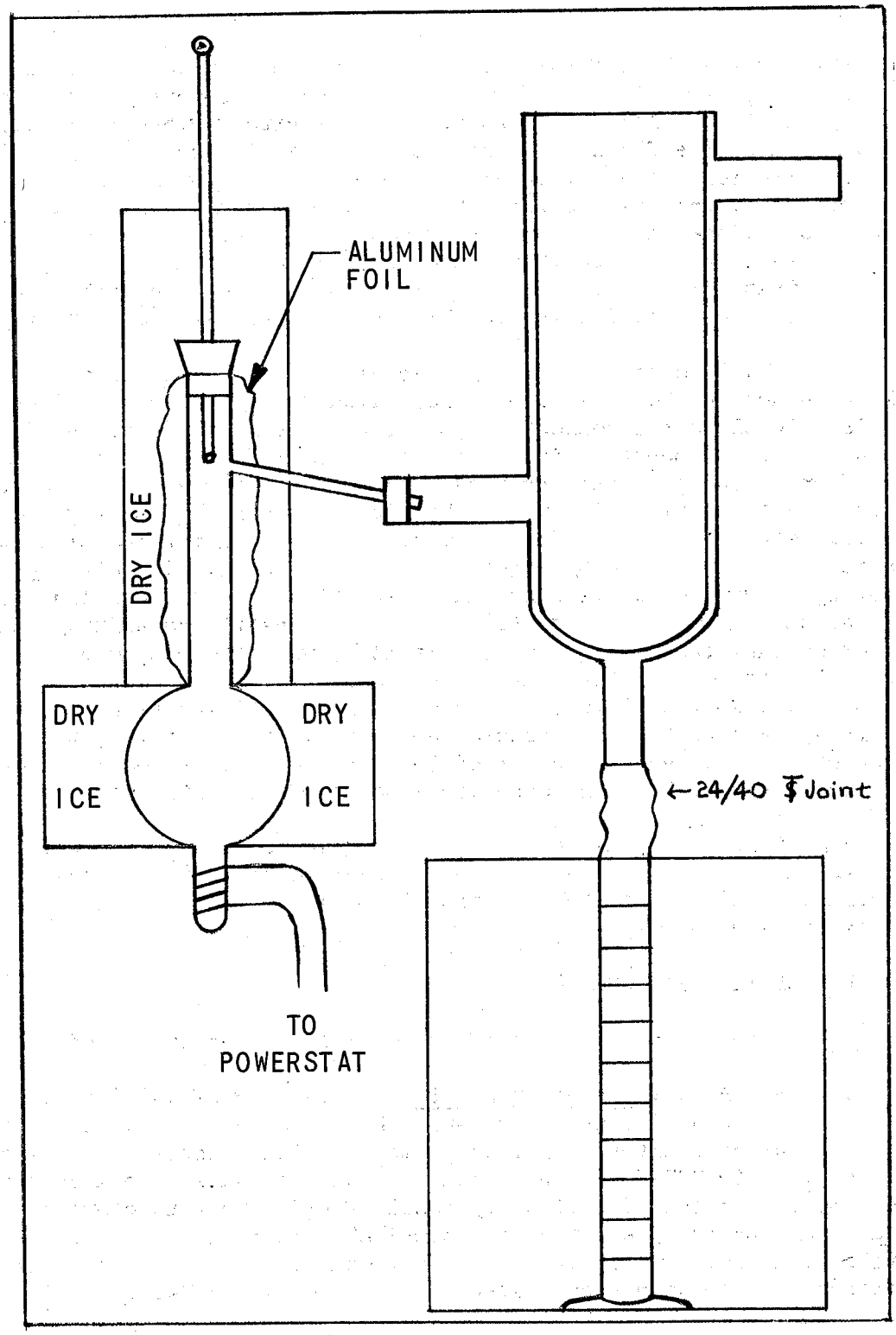


Figure 1. Distilling Apparatus

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holds the dry ice. The flask fits in the innermost space with about 1 cm of air space all around. This compartment is covered with a circular lid made from 1/32-inch gasket material with a 3.5-cm hole to fit the neck of the flask. The neck of the flask is surrounded by a piece of wire mesh 11 cm by 14 cm and covered with aluminum foil to prevent contact of the dry ice with the flask. The dry ice is held in place by a cylinder 10 cm in diameter with a slit cut in the side for the delivery tube. The delivery tube is fitted to an acetone-dry ice condenser that delivers to a 100-ml graduate in a wet ice bath. An ASTM No. 33C thermometer conforming to ASTM E1 is fitted with a cork stopper.

(b) Procedure. Cool the specimen in a dry ice-acetone bath. Fill the space in the lower compartment with dry ice. Set the distilling flask in place and connect it to the condenser. Set the outer cylinder in place, stopper the distilling flask, and fill the space with dry ice. Connect the graduated cylinder to the condenser and fill the beaker surrounding this cylinder with crushed ice. Attach the heating element to a powerstat. Cool another 100-ml graduate. Transfer 100 ml of the specimen to the graduate by means of a powder funnel. Immediately remove the stopper from the distilling flask, insert a long stemmed funnel, and pour the specimen into the distilling flask. Insert and adjust the thermometer so that the top of the bulb is level with the bottom of the delivery tube exit. As soon as the thermometer reads -10° C turn on and adjust the powerstat so that 4 to 5 ml per minute is distilled (approximately 70 volts). Record the temperature when the first drop falls, then at 1 ml and at 95 ml. Correct the observed temperatures for barometric pressure. Determine the distillation range for 1 to 95 ml of distillate.

4.2.4.4 Container leakage. Completely immerse the unit container in water which has been preheated to 55° C until equilibrium is reached. A continuous emission of bubbles from the unit container shall be considered evidence of leakage.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A, 1-1/2-pound quantity. A quantity of 1-1/2 pounds (lb) (+1/4 or -0 ounce) of insecticide shall be packaged in a metal can in accordance with Department of Transportation (DOT) regulations. The can shall conform to DOT Specification 2P. Each can shall also meet minimum requirements for type IX, class optional of PPP-C-96 with commercial designation number 25 electrolytic tin plate and plan B exterior protective coating with side seam striped as specified therein. The can shall show no evidence of leakage when tested as specified in 4.2.4.4.

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5.1.2 Level C. A quantity of 1-1/2 lb (+1/4 or -0 ounce) or 150 (+1.5 or -0) lb as specified (see 6.2) of insecticide shall be packaged in accordance with DOT regulations and in a manner to assure integrity of the container without alteration of contents. The container shall protect the contents from the supply source to the first receiving activity and for a limited time of storage.

5.2 Packing. Packing shall be level A or C as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 One and one-half-lb quantity. Twenty-four 1-1/2-lb cans of insecticide, packaged as specified in 5.1.1, shall be packed in two tiers of 12 cans per tier in a three by four can arrangement in a close-fitting fiberboard box conforming to PPP-B-636. The tiers shall be separated by a fiberboard pad. Movement of the cans within the box shall be prevented by the insertion, where needed, of fiberboard pads. Fiberboard pads shall be formed from material conforming to type CF, class domestic, variety optional, minimum grade 125 of PPP-F-320. The fiberboard box shall be overpacked in a close-fitting wood box for a type 2 load conforming to class 3, style 2 or 3 of PPP-B-585; overseas type, style A, B, or I, grade A of PPP-B-601; or class 2, style 4, grade A of PPP-B-621. Movement of the fiberboard box within the wood box shall be prevented by the insertion, where needed, of fiberboard pads conforming to minimum grade W5c of PPP-F-320. The wood box shall be closed and reinforced by strapping as specified in the appendix to the applicable box specification.

5.2.1.2 One hundred fifty-lb quantity. A quantity of 150 (+1.5 or -0) lb of insecticide shall be packed in accordance with DOT regulations in a metal cylinder conforming to DOT Specification 3AA. The cylinder shall also conform to the applicable requirements of RR-C-901, type II, class 1. The cylinder shall show no evidence of leakage when tested as specified in 4.2.4.4.

5.2.2 Level C. Twenty-four 1-1/2-lb cans or a 150-lb quantity as specified (see 6.2) of insecticide, packaged as specified in 5.1.2, shall be packed in a container which will insure safe delivery to destination. The container shall comply with DOT regulations and Uniform Freight Classification rules or National Motor Freight Classification rules, as applicable. Containers shall be sufficiently rigid to withstand transportation and to permit the containers to be stored in tiers at least 10 feet high without damage to containers or contents. The gross weight and size limitations of the container shall not exceed the tariff or classification limitations and the containers shall be closed or strapped as specified therein.

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5.3 Marking and labeling. Marking and labeling shall be in accordance with DOT regulations and Fed. Std. No. 123 for civil agencies or MIL-STD-129 for military activities. The label shall comply with the Federal Environmental Pesticide Control Act and shall be accepted by and registered with the Environmental Protection Agency. Each unit container shall be labeled to include precautionary marking, first aid, directions for use as a fumigant, active ingredient information, and lot number. Labels for military procurements shall include the following statement: "FOR MILITARY USE ONLY - NOT FOR RESALE UNDER THIS LABEL." The precautionary and first aid labeling shall include the following information:

DANGER! POISON
KEEP OUT OF THE REACH OF CHILDREN
HIGHLY VOLATILE
CAUSES BURNS

Poisonous liquid and vapor.
Do not breathe vapor.
Do not get in eyes, on skin, or on clothing.
Wear a full face gas mask with black canister meeting the U.S. Bureau of Mines specification for organic vapors.
Keep away from heat.
Avoid direct contact with foam rubber, leather, or wool.
This can is under pressure and should be opened only with special device intended for this purpose.*

FIRST AID TREATMENT - ANTIDOTE

Send for doctor. Place patient in fresh air, face downward, with head slightly below level of lungs. Keep warm. Give artificial respiration if breathing has stopped. Use oxygen inhalator or caffeine stimulants only at the direction of a physician.

In case of contact, immediately remove all contaminated clothing, shoes, or bandages, and wash skin thoroughly with soap and water. Flush eyes with water for at least 15 minutes. Do not re-use clothing or shoes until free from all contamination.

Do not enter treated space until thoroughly ventilated and tested safe with a suitable halide leak detector. If necessary to enter treated space without testing, use a Government approved mask (organic vapor mask or compressed air or oxygen generating breathing apparatus).

*Omit this sentence on cylinders.

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5.4 Palletization. When specified in the contract or order (see 6.2), uniform quantities of shipping containers shall be palletized. For level A and level B protection, shipments shall be palletized in accordance with the applicable requirements for load type I of MIL-STD-147 for shipping boxes and in accordance with load type IX of MIL-STD-147 for cylinders. The pallet shall conform to style 1, type I, class 1 of MIL-P-15011. For level A shipments, steel strapping shall be used and pallets and wood storage aids shall be treated with preservative as specified in MIL-P-15011. For level B shipments, steel strapping shall be used and pallets and wood storage aids shall not be treated with preservative. For level C shipments, palletization shall conform to DOT regulations, Uniform Freight Classification rules, or National Motor Freight Classification rules, as applicable. The palletization shall permit stacking of palletized shipping containers in tiers at least 10 feet high without damage.

6. NOTES

6.1 Intended use. Insecticide, methyl bromide is intended for use as a general fumigant for household effects, certain food and clothing products, and soil.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type of insecticide required (see 1.2).
(c) Level of packaging and packing required (see 5.1 and 5.2).
(d) Unit quantity required (see 5.1.2 and 5.2.2).
(e) If palletization is required (see 5.4).

6.3 Batch. A batch is defined as that quantity of material which has been manufactured by some unit chemical process or subjected to some physical mixing operation intended to make the final product substantially uniform.

6.4 Significant places. For the purpose of determining conformance with this specification, an observed or calculated value should be rounded off "to the nearest unit" in the last right-hand place of figures used in expressing the limiting value, in accordance with the rounding-off method of the Recommended Practices for Designating Significant Places in Specified Limiting Values (ASTM E29).

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MILITARY INTEREST:

Custodians:

Army - EA
Navy - SH
Air Force - 68

Review activities:

Army - GL, MD
DSA - GS

Preparing activity:

Army - EA

CIVIL AGENCY COORDINATING ACTIVITIES:

AGR - ARS
EPA - PRD
GSA - FSS
VA - DMS

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

Insecticide, Methyl Bromide O-I-556C

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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